

Valve Tests 2

Written by WB, 22/07/2026.

Summary

This repository is intended to replace the various scripts written by HL for the 7.9mm valve test rig. It comprises a polling script, which listens for user inputs via the nanodac, and a number of functions for automating the test procedure.

Usage

Testing valves

Make sure to run the polling script before beginning a test. To run the program, double click *Run Valve Tests.bat*.

To execute a function, change User Value 1 on the nanodac to the corresponding number. The polling function is latched, so the function will run only once then wait until the value is reset to 0.

Available functions:

Number	Function
1	Test & record cracking pressure

Settings

To change the program settings, open the file `settings.json` in a text editor.

Controller & MODBUS addresses

Hardware addresses are stored in the following structure:

```
addresses
├─ controllers
│   └─ "name A": "xxx.xxx.xxx.xxx:xxxxx"
│   └─ "name B": "xxx.xxx.xxx.xxx:xxxxx"
```

```

|   └─ etc...
└─ modbus
    └─ programmer
        |   └─ "UserInput":  ["controller name", "modbus address"]
        |   └─ "TempSensor": ["controller name", "modbus address"]
        |   └─ "RampControl": ["controller name", "modbus address"]
    └─ pressure
        |   └─ "ColNameA": ["controller name", "modbus address"]
        |   └─ "ColNameB": ["controller name", "modbus address"]
        |   └─ etc...

```

Note that "etc..." indicates an arbitrary number of entries. Ensure "controller name" fields match the controller name exactly.

File handling

By default, files are saved in the format `Documents/ValveTests2/YYYY-MM-DDThh-mm-ss.csv` (based roughly on ISO 8601).

The available settings are listed below.

Setting	Code	Notes
Save file location	<code>{"SavePath": "C:\path\to\folder\}"</code>	Paste file path from Windows console to avoid formatting errors
Save file name convention	<code>{"FileName": "results"}</code>	A date or time value can be specified according to strftime rules ; if a plaintext filename is used, files will be named sequentially rather than timestamped (i.e. NewFile 1, NewFile 2, etc.)